

Venkata Kishan Kapuganti

Dallas ,TX | (469) 726-9026 | Venkata.kishan.vk2@gmail.com | [linkedin](#)

Summary

Data Engineer with proven expertise in scalable cloud pipelines and automation. Experience in Linux scripting, AWS operations, and application monitoring using tools like Datadog. Adept at supporting high-availability production systems while collaborating with cross-functional teams to enhance uptime and reliability. Committed to leveraging technical skills to drive business continuity and operational excellence.

Skills

- **Programming & Scripting:** Python, SQL, Shell Scripting
- **Data Engineering:** ETL/ELT Development, Data Ingestion, Data Modeling, Data Warehousing, Workflow Orchestration
- **Big Data Tools:** Apache Airflow, Databricks, Alteryx
- **Databases & Storage:** Snowflake, AWS S3, AWS Redshift, Oracle, MySQL, PostgreSQL, MongoDB
- **Cloud Platforms:** AWS (S3, Redshift, EC2, CloudWatch), Azure (basic)
- **DevOps / CI-CD:** Jenkins, Git, Bitbucket, Docker, Kubernetes, CI/CD Pipelines for SQL and Python
- **Monitoring & Automation:** AWS CloudWatch, ServiceNo
- **Visualization Tools:** Tableau, Matplotlib, Seaborn
- **APIs & Microservices:** REST API Integration
- **Tools & Project Management:** JIRA, Confluence
- **Methodologies:** Agile, Scrum (Sprint Planning, Standups, Retrospectives)
- **System Administration & Security:** Linux, System Administration, Infrastructure Security

Experience

Citius Tech USA

Feb 2024 - Present

Data Engineer

- Developed robust ETL pipelines using Python, Java, and SQL to process high-volume structured and semi-structured data from multiple systems, ensuring data integrity and compliance with industry standards.
- Automated and scheduled 90+ Airflow DAGs daily, enabling real-time analytics and eliminating manual data ingestion processes.
- Integrated Kafka with PySpark-based streaming pipelines to ingest and process 500K+ patient vitals daily, reducing real-time alert latency by 60% and supporting rapid clinical interventions.
- Containerized Python services using Docker and deployed via Jenkins CI/CD pipelines, reducing deployment errors by 90% and cutting release cycles by 60%.
- Integrated multiple third-party REST APIs to enrich patient records, reducing data unavailability for care teams by 2 hours on average.
- Implemented real-time data monitoring with AWS CloudWatch and set up 15+ automated alerts for schema mismatches, pipeline failures, and data integrity issues, reducing incident detection time by 40%.
- Automated Alteryx workflows for patient data cleansing, cutting manual effort by 35% and enhancing SLA compliance.
- Collaborated with visualization teams to deliver business intelligence dashboards using Tableau, supporting operational insights and informed decision-making.
- Enforced Snowflake row-level security and column masking across 12+ user roles to ensure 100% HIPAA compliance.
- Reduced healthcare data delivery time from 8 hours to under 30 minutes, enabling faster decisions for 1,000+ patient records daily.
- Coordinated with data analysts, QA engineers, and DevOps teams via JIRA and Confluence to deliver high-quality, production-grade data products, enhancing overall data processing efficiency and system reliability.

Tata Consultancy Services India

Sep 2019 - Dec 2022

System Engineer

- Designed and deployed 50+ SQL-based ETL workflows processing over 2TB of monthly telecom data into Redshift and reporting stores.
- Automated job status checklists for 10+ teams, reducing manual reporting time by 80% and improving operational transparency.
- Diagnosed and resolved 200+ job failures, boosting data pipeline reliability to 99.2% through Jenkins and ServiceNow integrations.
- Utilized Databricks notebooks to optimize ETL workflows and machine learning experiments, reducing model training time by 30% on telecom datasets.
- Developed a Kafka-based ingestion prototype to stream 100K+ daily telecom device logs into monitoring systems, automating error detection workflows and decreasing manual log review by 70%.
- Resolved 150+ support tickets with 95% SLA adherence by performing in-depth root cause analysis and implementing sustainable fixes.
- Constructed CI/CD pipelines for SQL deployment using Jenkins and Git, facilitating faster development cycles and reducing manual release errors.
- Designed 10+ Tableau dashboards for monitoring SLA, device metrics, and performance, supporting real-time reporting for 30+ stakeholders.

- Leveraged AWS services, including EC2, S3, and Redshift, to build scalable and secure cloud-based data processing and storage solutions.
- Participated in Agile sprints through daily stand-ups, sprint planning, and retrospectives to ensure continuous progress and alignment.

Hexaware Technologies India

May 2019 - Aug 2019

Machine Learning Intern

- Developed 4 supervised machine learning models with 85–92% accuracy using Scikit-learn for both classification and regression tasks.
- Conducted comprehensive data preprocessing including missing value treatment, feature scaling, and encoding to prepare datasets for effective model training.
- Assessed model performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.
- Created 15+ visualizations using Matplotlib and Seaborn for stakeholder presentations and model validation.

Education

University of North Texas- Denton, TX

Aug 2024

Masters, Computer and Information Sciences

Gandhi Institute of Technology, Andhra Pradesh, India

2019

Bachelors, Computer Science Engineering

Projects

Heart Disease Prediction using Machine Learning

Jan 2023 - Apr 2023

University of North Texas

United States

- Developed a machine learning model using Python and Scikit-learn (Logistic Regression, Random Forest) to predict heart disease with 90%+ accuracy.
- Handled complete data preparation workflow including preprocessing, feature selection, and evaluation using F1-score and ROC-AUC metrics.

Enterprise Order Management System

Jan 2024 - Apr 2024

University of North Texas

United States

- Created a Java-based application with MySQL backend to manage customer orders, inventory, and dispatch, simulating a small enterprise ERP system.
- Designed and executed SQL queries for CRUD operations and reporting; integrated database using JDBC and applied OOP principles for backend logic.

Certifications

- AWS Certified Developer - Associate
- Oracle Certified Java Programmer

Achievements

- **Top Performance Band “A” at TCS::** Recognized with the highest performance rating band for delivering consistent backend solutions, automation, and CI/CD enhancements across telecom-based enterprise systems.
- **5/5 First-Year Appraisal Rating:** Earned a perfect rating in performance evaluation during the first year at TCS for strong ownership in portal development, API security, and client satisfaction.
- **Reduced Real-Time Alert Latency by 60%:** By integrating Kafka with PySpark streaming, enabled faster clinical decision-making and reduced patient risk in critical care environments.
- **Achieved 100% HIPAA Compliance:** Designed and implemented Snowflake security controls, including row-level security and column masking across multiple user roles.